

Name: Sonia Yi

Internship or thesis: Thesis

Title Project: Cerebellar Mechanisms in Reward Anticipation and Outcome:
The Role of Motivational Traits

Abstract:

Reward anticipation and outcome processing are central components of motivated behavior and adaptive learning. Although the cerebellum has traditionally been associated with motor control, growing evidence suggests that it also contributes to reward-related and predictive cognitive processes. However, the extent to which cerebellar involvement in reward processing is modulated by individual differences in motivational traits remains unclear. The present study aims to investigate cerebellar mechanisms underlying reward anticipation and outcome processing, and to examine how motivational traits influence these neural responses. Healthy participants will complete a reward anticipation-reward omission task while EEG is recorded. Neural markers associated with reward expectation, outcome evaluation, and prediction error will be examined. Individual differences in motivational traits, including rewards sensitivity and approach motivation, will be assessed using the Behavioral Inhibition System/Behavioral Activation System, (BIS/BAS) questionnaire. Primary analyses will focus on associations between motivational traits and neural responses during reward anticipation and outcome.

Name of company or university: Utrecht University

UU Supervisor: Dennis J.L.G. Schutter

Name Student: Markus Kaiser

Internship or Thesis: Internship

Title project: Creating a meaningful first user experience in VR

Name company or University: Haptive

(If applicable) External supervisor: Alexander Achberger

UU Supervisor: Samson Chota

Abstract:

VR can be used for a multitude of applications. Besides the obvious use for entertainment purposes it is also often used for serious topics like trainings, simulation or creation of prototypes. In each of the use-cases a major issue is that you can only replicate the visual world and have no physical feedback. This problem is partly solved through the STRIVE, a string based haptic feedback system developed by HAPTIVE. For this system I am creating and comparing 3 different first user experience scenarios in which the user is guided through the installation of the system. The goal is to create a fully immersive instruction for any user. This VR experience is created using the Unity software and C# script. The tutorial will be tested and improved through usability testing and understanding. Qualitative data will consist of statements gathered through think aloud method and the SUS filled out by testers. Lastly The project is also about supporting during fares and creating use-cases to present the system.

Name Student: Léon de Vet

Internship or Thesis: Thesis

Title project: The problem of data-loss and participant exclusion in eye-tracking research: A pilot study into the positive skew

In eye-tracking research a large portion of participant data is lost due to inaccuracies in measurements. This multi-staged study will investigate potential causes and solutions for this problem. The study is currently in the pilot phase. First the study will introduce a perceptual task. In this perceptual task the participant is first shown a fixation point followed by the stimulus, a 'C' with the gap pointing in four different directions (left, right, up and down) for a certain period of time, followed by a mask. The participant must then report the direction of the stimulus that was presented. In the first pilot a ceiling effect was reached, and no effect was found for gap size. In the second pilot flankers were added to the stimulus and the mask was made more disruptive, and the presentation time was shortened. In the second pilot a floor effect was reached.

Name company or University: Utrecht University

UU Supervisor: Roy Hessels

Name Student:

Fotios-Konstantinos Perikleous

Internship or Thesis:

Internship

Title project:

Development and Launch Preparation of an AI Assistant for Collaborative Efficiency

Abstract:

This internship at CoVince began as an internal experiment aimed at exploring how an AI assistant could improve efficiency in group and collaborative settings. Over time, the project evolved into the structuring, optimization, and launch preparation of a market-ready AI assistant product. My role focused on three main areas: conducting a literature review on prompt engineering and conversational AI behavior, testing and refining prompts to enhance the assistant's alignment, consistency, and performance, and developing a marketing plan and visual promotional materials under supervisory guidance. Through systematic research and prompt testing, effective prompt structures were identified and directly informed the assistant's behavior and usability prior to release. In parallel, marketing strategies and visuals were developed to align with the product's positioning and target audience. This experience strengthened my research, analytical, product development, and communication skills while providing hands-on exposure to real-world AI product deployment.

Name company or University:

CoVince

External supervisor:

Richard van Tilborg

UU Supervisor:

Anouk Keizer

Name Student: Steven van Harselaar

Internship or Thesis: Internship

Title project: Pupil Dilation Reflects Auditory Working Memory Load in Visually Naturalistic Settings

Pupil size is often used as a powerful marker of mental effort, but its practicality in more natural settings is unclear because pupil size is also affected by low-level visual features such as luminance. In this study, we investigated whether pupil size reflects task load in both controlled and more naturalistic conditions. Participants performed an auditory 0-back, 1-back, and 2-back task while either watching a video of a car drive or viewing a fixation cross. Task performance, subjective workload (NASA-TLX), and pupil size were measured. Performance decreased, while workload, and pupil dilation increased for the 2-back condition compared to the 0-back and 1-back conditions, which did not differ from each other. The effect of n-back level on pupil dilation was similar in both movie and no-movie conditions. These findings suggest that pupil dilation may be able to reflect auditory task load even in visually more complex environments, despite low-level visual features.

Name company or University: Utrecht University

UU Supervisor: C. Strauch & C. Yuqing

From Obligation to Engagement: Design Principles for AI Literacy E-learning in Government Organizations

This article examines which design and communication principles motivate employees at SSC-ICT to voluntarily participate in e-learning modules on AI literacy. The rapid development of Artificial Intelligence (AI), combined with the requirements of Article 4 of the European AI Act, obliges organizations to ensure a sufficient level of AI literacy among all staff. This poses a particular challenge for organizations with a highly heterogeneous workforce, ranging from technical experts to non-technical policy advisors.

A literature review based on Self-Determination Theory and the Technology Acceptance Model identified 13 relevant design principles, explored through six in-depth interviews. These are synthesized into five overarching design strategies: (1) an expert track via diagnostic entry assessments, (2) role-specific, value-based communication of usefulness, (3) short, mobile-accessible micro-learning to ensure operational fit, (4) positioning AI literacy as part of a shared public mission, and (5) a growth-oriented learning environment without controlling mechanisms. By prioritizing perceived usefulness and autonomy, AI literacy can shift from a compliance requirement to a professional development.

Key words: AI literacy, motivation, e-learning, Self-Determination Theory, Technology Acceptance Model

Name Student: Anna Sas (4903641)

Internship or Thesis: Internship

Title project: An Exploration of Opportunities to Engage Young Adults in the Repositioning of VluchtelingenWerk Nederland

The internship focused on exploring the opportunities to engage young adults with the organisation VluchtelingenWerk Nederland (VWN). As of January 2026, VWN will be implementing a repositioning strategy that aims to redefine the organisation's position within society and the way of communicating to the public. This renewed message aims to engage a broader audience and focusses on the "movable middle". Within this context, the project examined to identify opportunities to engage young adults within the context of VWN's repositioning. These opportunities were explored through a review of academic literature, the objectives and meanings of the repositioning strategy, and a review of other non-governmental organizations (NGOs). The findings highlight the importance of targeting young adults, as this group can contribute positively to both the short- and long-term impact of VWN. To effectively engage this audience, new communication strategies are required to be implemented which provide long-term connection and involvement with the organisation.

Name company or University: VluchtelingenWerk Nederland

External supervisor: Robin Siep

UU Supervisor: Gabriël Beckers

Student: Sander de Graaf

Supervisor: Marnix Naber

Thesis

Utrecht University

Acoustic Predictors of Pupillary Response During Music Listening Under Simulated Hearing Loss

This project investigates how musical sound features predict pupillary responses during listening under normal and reduced hearing. Pupillometry provides an objective measure of listening effort, but the contribution of specific acoustic properties is still unclear. Twenty normal-hearing adults listen to short clips from ten genres under four conditions: intact, moderately degraded, strongly degraded, and intact music over high-frequency-attenuating earplugs. Pupil size is continuously recorded with an EyeLink tracker, while time-varying features such as loudness, spectral content, and timing are extracted with Librosa. A pupil response function is used to model how pupils track brief changes in these features, revealing how coupling between music and pupil size changes with simulated sensorineural loss and spectral attenuation. The study aims to disentangle feature-driven from effort-related pupil responses and to inform the design of engaging, music-based hearing assessments.

Name Student: Tijmen van Kooten

Internship or Thesis: Thesis

Title project: Exploring implicit visual attention as a measure of sexual orientation.

Abstract project:

In this thesis, I investigate whether a new implicit eye-tracking task can detect differences in sexual orientation through visual attention and reaction times. For socially sensitive topics, explicit questionnaires can be influenced by social desirability, which may obscure underlying preferences. To address this, we use a rapid gaze-shift task in which participants respond to a color cue after a very brief (30 ms) presentation of an image of a naked person. We expect that stimuli matching a participant's sexual preference will hold attention longer, leading to slower gaze-shift reaction times. To validate this method, participants also complete an explicit eye-tracking task with longer stimulus exposure (200 ms), an Implicit Association Test (IAT), and a short questionnaire. The study consists of one session of approximately 60 minutes. Some stimuli include AI-generated nude bodies.

Name company or University: Universiteit Utrecht

UU Supervisor: David Terburg

Name Student: Sophie Oostenbruggen

Internship or Thesis: Internship

Title project: Optimizing Neutralization Time in Sensory Testing: Effects on Product Discriminability and Data Quality in a Dairy CLT

Name company: Essensor

External supervisor: Daan Schijvens

UU Supervisor: Monique Smeets

This study examined whether shortening time per product (and thus available neutralization time) affects product discriminability and the quality of sensory and acceptance evaluations in a Central Location Test (CLT). Seventy participants evaluated four forest fruit yoghurts under one of three between-subject time-per-product conditions: 9 minutes (Control; $n = 29$), 6 minutes (Treatment 1; $n = 20$), or 3 minutes (Treatment 2; $n = 21$). Sensory attributes were rated using 100-mm VAS scales, complemented by control questions and a final questionnaire assessing perceived neutralization time, stress, carry-over, and discriminability. Data were analyzed using linear mixed-effects models. No group $\times$ product interactions were found, indicating that product discriminability did not depend on time per product. Reducing time per product from 9 to 6 minutes had no adverse effects on sensory or liking ratings. In contrast, the 3-minute condition increased perceived time pressure and reports of insufficient time without clear changes in attribute means. These findings suggest that a 6-minute time per product offers an efficient and reliable option for similar dairy CLTs.

Name student: Thijme Rigter

Internship: The World Is Your Lab, Utrecht University

UU supervisors: Stella Donker and Roy Hessels

The Adaptation of Digital Learning Platforms: Brightspace Pulse at Utrecht University

Abstract

This study focuses on the factors that contribute to the adaptation of the digital learning platforms of Utrecht University, especially (Brightspace) Pulse, and how to improve its usage and awareness. The Pulse app seems to lack in those factors as a mobile app, despite it having the same functionality and course related support as the desktop version. For this study, a quantitative research method was conducted based on an online questionnaire (N=39) with adaptation of the System Usability Scale (SUS). To clarify this data, a qualitative interview method was used (N=5). Results indicate that the low awareness of Pulse was caused by low prominence, indistinct communication about its existence, and uncertain benefits to the student's study routine. Furthermore, the results demonstrate that usage rates could be enhanced by improved communication and engagement strategies. Considering these findings, the study recommends improved integration and clearer communication by Utrecht University about Pulse.

Key words: Brightspace Pulse, Utrecht University, Digital Learning Platform, System Usability Scale, MyUU

Name Student: Nicoline van Vught

Internship or Thesis: Internship

Title project: Mixed Reality in Healthcare: A Feasibility Study of Holomoves at the Diakonessenhuis

Abstract:

Method: A mixed-methods design was used; surveys with patients (n = 16) and physiotherapists (n = 6), semi-structured interviews with nursing staff and department management, and observational sessions. The goal was to identify exclusion criteria for patients to use MR in rehabilitation.

Results: Results indicated that patients were enthusiastic; however, independent use among patients aged 65+ was limited, and individuals who already had more than 180 minutes of daily physical activity were less motivated to participate. Physiotherapists enjoyed using the headset, but noted barriers such as setup time and selecting fitting patients. Nurses showed enthusiasm and viewed MR rehabilitation as part of their responsibilities.

Conclusion: Based on the observations and interviews, patients with cognitive impairments, movement restrictions, or attached medical lines were excluded for their own safety, while those with conditions such as FND were excluded due to the high level of supervision required. Patients without these limitations represent the most suitable target group for MR rehabilitation, particularly hospitalized patients who require additional motivation to remain active.

Name company or University: Holomoves

External supervisor: Joep Janssen

UU Supervisor: Larissa Meijer

The effect of sport activity on body perception and the moderation influence of muscle dysmorphia

Body image has predominantly been conceptualized in visual and cognitive terms, while considerably less attention has been paid to body perception, defined as the somatosensory representation of the body based on tactile and proprioceptive input. Emerging evidence suggests that body image disturbances are not limited to visual evaluations but also involve systematic distortions in tactile body size perception. This reflects alterations at a relatively automatic and embodied level of body representation rather than conscious cognitive processes.

Sport activity may be associated with such somatosensory distortions, particularly when accompanied by elevated muscle dysmorphia-related attitudes, which are characterized by a preoccupation with perceived insufficient muscularity and increased body-focused attention. The present study examined the relation between sport activity and tactile body perception, with muscle dysmorphia-related attitudes as a potential moderating factor.

Using a cross-sectional design, young adults aged 18–30 years ($N \approx 76$) completed self-report measures of sport activity (Baecke Habitual Physical Activity Questionnaire), muscle dysmorphia-related attitudes (Swansea Muscularity Attitudes Questionnaire), and body appreciation (BAS-2). Somatosensory body perception was assessed using a Tactile Estimation Task (TET), alongside a Visual Size Estimation task (VSE).

Results indicated that higher levels of sport activity were associated with greater systematic tactile body size underestimation. This relationship was moderated by muscle dysmorphia-related attitudes, such that the association was strongest among individuals with elevated muscularity-related concerns. No comparable effects were observed for visual body size perception. These findings highlight the importance of considering embodied processes in research on body image disturbances.

Name Student: Muhammad Danish Khoo

Student number: 8466017

Internship/Thesis: Internship

Buffering negative emotions with attribution: An observational study of failure in pinball play

We investigate how attribution practices can help relieve the negative emotions, such as valence, arousal, and control, felt from experiencing failure in games. Player behaviour in games (video and physical) can be explained by the paradox of failure, where the player seeks out failure despite the negative feelings. What is not known, however, is the strategies to attenuate these negative feelings to promote a healthier form of play. The following study expands on this paradox in a within-subject observational study done across 2 on-site festivals concerning pinball machines. The results were analysed using Python, where we found that attribution theory can buffer negative feelings associated with failure through a 2-step appraisal process. Additionally, we find a trend supporting the notion that subjective performance indicators, such as time played, can be a stronger factor in getting players to take responsibility for their failures, thereby turning the failure into a correctable and, more importantly, less negative experience. Taken together, the study provides insight into understanding player behaviour and promoting a healthier form of play.

Company/University: Utrecht University, Centre for Game Research

External supervisor: Sander Bakkes (s.c.j.bakkes@uu.nl)

UU Supervisor: Leendert van Maanen (l.vanmaanen@uu.nl)

Name Student: Manon Middendorp

Internship or Thesis: Internship

Title project: Assessing Implementation Prospects of Personalized Learning Programs for Career Changers entering the Dutch Armed Forces

Abstract:

The Dutch Armed Forces are required to achieve a fast-tracked increase in employment which requires the basic military training (BMT) to get optimized. Career changers, entering with experience and skills, follow a fast-track BMT. Leveraging their competencies through personalized learning might be beneficial in optimizing their learning process. By conducting semi-structured interviews with career changers who completed the BMT, this study assessed implementation prospects of personalized learning in the BMT. Results showed that possibilities lie in personalizing the BMT to physical fitness, guiding self-reflection and providing purpose of specific learning content. Moreover, interviews revealed that time and group-focused culture were limiting factors of implementing personalized learning in the BMT. Based on these findings, it is suggested that relatively low-fit career changers follow a preparatory physical training. Additionally, a Large Language Model could be designed to guide self-reflection and provide purpose of learning content related to career changers' expected future positions.

Name company or University: TNO: Innovation for life

External supervisor: Pauline Zuidema

UU Supervisor: Chris Paffen

Student name: Enise Hidir
Thesis: Numerosity
Organization name: Utrecht University
Assessor: Maarten van der Smagt
Auditor: Samson Chota
Student number: 7240597

Exploring the Role of Contrast Polarity in Numerosity Adaptation

The perception of numerosity, or the ability to estimate the number of items in a visual array, is a fundamental cognitive skill that supports everyday behaviour and higher-level cognitive processes. Numerosity perception is influenced by several visual properties, including spatial arrangement, and luminance. Contrast polarity provides a useful way to examine where numerosity is processed within the visual system, as positive and negative contrast are processed through separate pathways up to early visual cortex. In this study a numerosity adaptation task will be used to investigate whether numerosity adaptation depends on contrast polarity. Participants will complete a within-subjects experiment involving dark and light dot arrays presented across eight conditions and make comparative numerosity judgments. These data will be analysed with the help of a repeated measures ANOVA.

Word count (excl. title): 127

Name Student: Niek Ypma (7261527)

Internship or Thesis: Internship

Title Project: *"Our input is valuable": End Users' Goals, Needs, and Perspectives on Change Management in the Design of New Public Transport Fleet and Depot Management Applications.*

This internship examines how vehicle controllers' tasks and needs at Qbuzz can inform application design and change management for a new fleet- and depot-management system. In the current application development process, end users experience change fatigue, limited access to information, and insufficient involvement, resulting in frustration, diminished trust, and reluctance to provide feedback, undermining development efficiency and adoption potential. Through semi-structured interviews with vehicle controllers (N = 7) and literature review, the study identifies task requirements, three core functionalities: garage parking overview, bus deployment screen, and Service & Logistics deployment screen, and vision on change management. Findings emphasize that successful adoption requires not only appropriate technical features but also clear communication, timely feedback cycles, and accessible training throughout development. By centering end users in the process, this research demonstrates that meaningful involvement and transparent communication are essential to building trust and ensuring successful technology development and implementation in organizational contexts.

(150 words)

Name of the company: Qbuzz in collaboration with The World is Your Lab

External Supervisor: Lisa Theuns-Braam & Daniëlle Dielemans

UU Supervisor: Roy Hessels

Name student: Daan van der Wees

Internship or Thesis: Internship

Title Project: Finding the Sweet Spot: A Neuromarketing Study of Sweets in Supermarkets

In supermarkets, brands and products are constantly competing for your attention. In this specific case, a Dutch supermarket chain was redesigning their sweets aisle. For this study, a sweets company needed to know what the best place for them on the shelf was. They used our report to negotiate a favorable place in the sweets aisle in the supermarket. Participants completed a computer study viewing different supermarket sweets-shelves, with varying assigned goals, ranging from complete freedom to targeted searches. Data was collected using EEG and eyetracking technology. Data was analyzed in iMotions using eyetracking AOIs and EEG metrics. Three different top-performing shelves were found across different categories.

Name company or University: Unravel Research

(If applicable) External supervisor: Tom van Bommel

UU Supervisor: David Terburg

Enhancing CPR training: Recommendations for personalized approaches

Astrid de Vries (6727255)

Applied Cognitive Psychology

Internship 12,5 ECTS

Institute: TNO

UU supervisor: Chris Paffen

TNO supervisor: Anna Hulshof

70% of cardiac arrests occur outside of hospital settings. Completing a cardiopulmonary resuscitation (CPR) course contributes to public safety, and better preparedness in crisis situations. Personalized approaches in CPR courses can enhance CPR skills by adapting the training to learners' individual needs and preferences. This study examines the extent to which personalized methods are currently incorporated into the Red Cross CPR refresher course and proposes recommendations for improvement. A self-developed observational checklist was used to assess the presence and quality of personalization in didactic methods (e.g., learning analytics, personalization in tasks/feedback/learning path/instruction, blended learning, and learner autonomy) as well as non-technical first aid competencies (e.g., collaboration, communication, decision-making and leadership) using a mixed methods approach. Findings indicate that personalization is present but limited and that personalization mainly occurs informally or reactively. Based on these findings, recommendations are provided to systematically incorporate personalization in CPR training. Future research could use objective measures of learning outcomes to evaluate the effectiveness of the recommendations.

Keywords

Personalized learning – Adaptive learning – First aid education – Non-technical skills – Didactic methods – Observational study

Name Student: Ya-Lei Loth

Internship

Title project: Optimizing the Clearance Interface in ATC Radar Systems

Abstract: Air Traffic Control (ATC) radar systems are essential tools used by Air Traffic Controllers (ATCOs) to manage aircraft movements safely and efficiently. The clearance menu is a critical component, enabling controllers to issue instructions and maintain situational awareness. However, its current design may lack optimal structural clarity and visual efficiency, potentially affecting performance and adaptability. This project aims to improve the clearance menu through targeted, low-impact design changes that enhance usability and visual clarity while maintaining existing interaction patterns. The first steps included literature review, focusing on human factors, cognitive ergonomic and ATC interface design principles. This was then followed by a detailed analysis of the current clearance menu, including examining structural organization, lay-out, mouse warping and button sequences. To further broaden my understanding, qualitative interviews on the clearance were conducted with several ATCOs. Based on these findings, several mock-ups have been created through an iterative design process. If successful, this project may contribute to improvements of the ATC radar interface at MUAC.

Name company: EUROCONTROL Maastricht Upper Area Control Centre

External supervisor: Adam Tisza

UU Supervisor: Daatn van Rooij

Name Student: Hadeel Khawatmy

Internship or Thesis: Internship

Title project: Metacheck Report

Abstract:

MetaCheck is a web-based tool designed to support researchers in identifying common reporting and reproducibility issues in scientific manuscripts, either before or after publication. During this Master's internship, the focus shifted from the technical development of MetaCheck's analytical modules to the usability and clarity of its report output. At the start of the internship, the report mainly consisted of unpolished module outputs and raw tables, offering limited guidance for interpretation. The objective of this project was therefore to redesign the MetaCheck report from a user experience perspective, ensuring that results are communicated clearly and accessibly to researchers from diverse academic backgrounds.

The internship involved an iterative redesign of the report's structure, visual layout, language, and interaction elements, including summaries, navigation, tables, icons, and expandable content. User feedback was gathered through discussions with PhD students and researchers and used to guide successive refinements. The final outcome is an enhanced MetaCheck report.

Name company or University: Technical University of Eindhoven

External supervisor: Prof. Dr. Daniel Lakens

UU Supervisor: Dr. Sjoerd Stuit

Name Student: Nora Abid

Internship or Thesis: Thesis about numerosity

Title project: The influence of luminance contrast on perceived numerosity after adaptation.

Abstract, 146 words

Numerosity is a perceptual ability to estimate the number of items in a set. Numerosity perception can be influenced by adaptation, and luminance contrast has been shown to interfere with symbolic number processing. However, it remains unclear what role luminance contrast plays in non-symbolic numerosity perception. This project investigates whether luminance contrast affects numerosity adaptation. Participants complete a visual numerosity adaptation task, they will be exposed to dot arrays with either high or low numerosity combined with either high or low contrast. After adaptation, participants judge whether a test array or a reference array is more numerous. To these responses psychometric functions are fitted and points of subjective equality (PSEs) are extracted. These data will be analyzed with a repeated measures ANOVA. Different outcome patterns will clarify whether luminance contrast contributes to numerosity perception during adaptation or whether numerosity perception remains robust, affected only by adaptation.

Name company or University: Utrecht University

(If applicable) External supervisor: N/A

UU Supervisor: Maarten van der Smagt

Name Student: Ian Koopmans

Internship: Internship 12.5ECTS

Title project: Setting the Course for Standardized Situation Awareness Measurement in MARIN Simulators

The aim of this study is to find approaches to standardizing the measurement of Situational Awareness (SA) within MARIN's experiments. Current experimental practice relies on a mix of subjective and objective measures, which can lead to inter-rater variability and inconsistent interpretation across researchers. The aim of this work is to provide MARIN experimenters with a structured framework for selecting appropriate human-factors measurements based on the underlying research question, task context, and performance relevance. Drawing on established SA literature and best practices, the study distinguishes between eye-tracking, objective SA measures (e.g. freeze-probe techniques), subjective ratings, and performance metrics. The resulting framework supports more consistent experimental design and data interpretation. This research is relevant for the future of marine operations, as increasing system complexity, automation, and safety demands require robust and comparable human-performance assessments. The outcomes are directly applicable to MARIN researchers and contribute to improved validity and reliability in maritime human-factors experimentation.

Name company: MARIN (Marine Research Institute Netherlands)

External supervisor: Niklas van Duinen

UU Supervisor: Jana Klaus

Name Student: Michela Pittalis (0695335)

Internship Title: Measuring Leadership Behaviours in Utrecht University (UU) Psychology Students

Utrecht University: The World Is Your Lab

Supervisor: Roy Hessels

Abstract

This exploratory study extends the existing literature on leadership, specifically in higher-education student populations. This research was aimed at identifying patterns in students' self-reported leadership behaviours, as well as any differences between subgroups based on variables such as gender, level of study, and educational background. Data were collected via a Qualtrics survey from 70 psychology students at Utrecht University (UU) using a self-report tool known as the "Authentic Leadership Self-Assessment Questionnaire" (ALSAQ). The tool treats authentic leadership as a multi-layered construct with four main pillars: internalised moral perspective, relational transparency, self-awareness, and balanced processing. Results showed similar composite scores for the four pillars across participants, regardless of their level and year of study, programme track, age, gender, and educational background. Mann-Whitney U and Kruskal-Wallis statistical tests detected no significant differences across subgroups. These findings reveal consistent patterns in authentic leadership self-reports among UU psychology students across subgroups, highlighting the need for larger, longitudinal studies to explore subtle variations and inform educational programme development.

Keywords: authentic leadership, university students, self-assessment, leadership behaviours

Name Student: Mijntje Kranenburg

Internship or Thesis: Internship

Title project: Website Usability during Simulated Visual Impairment

In today's society, website usability and accessibility are critical. Despite existing guidelines for inclusive design, especially individuals with visual impairments continue to face challenges while navigating websites. This study aimed to address this issue by (1) systematically reviewing simulation tools for studying the impact of visual impairments on website usability and (2) examining how different simulated impairments affect users' task performance and experience. First, an overview of existing visual simulation tools was created. Next, participants completed web-based tasks under normal vision and experimental conditions simulating blurry vision and peripheral vision loss. Results showed longer completion times, higher subjective task load and lower perceived usability in the experimental conditions. These findings demonstrate that visual impairment simulations offer valuable insights into website usability and negatively affect both objective performance and subjective experience. The study underscores the importance of designing inclusive websites and provides a foundation for future research in this area.

Name company or University: Utrecht University (The World is Your Lab)

(If applicable) External supervisor: –

UU Supervisor: Roy Hessels

The effect of sport activity on body perception and the moderation influence of muscle dysmorphia

Body image has predominantly been conceptualized in visual and cognitive terms, while considerably less attention has been paid to body perception, defined as the somatosensory representation of the body based on tactile and proprioceptive input. Emerging evidence suggests that body image disturbances are not limited to visual evaluations but also involve systematic distortions in tactile body size perception. This reflects alterations at a relatively automatic and embodied level of body representation rather than conscious cognitive processes.

Sport activity may be associated with such somatosensory distortions, particularly when accompanied by elevated muscle dysmorphia-related attitudes, which are characterized by a preoccupation with perceived insufficient muscularity and increased body-focused attention. The present study examined the relation between sport activity and tactile body perception, with muscle dysmorphia-related attitudes as a potential moderating factor.

Using a cross-sectional design, young adults aged 18–30 years ($N \approx 76$) completed self-report measures of sport activity (Baecke Habitual Physical Activity Questionnaire), muscle dysmorphia-related attitudes (Swansea Muscularity Attitudes Questionnaire), and body appreciation (BAS-2). Somatosensory body perception was assessed using a Tactile Estimation Task (TET), alongside a Visual Size Estimation task (VSE).

Results indicated that higher levels of sport activity were associated with greater systematic tactile body size underestimation. This relationship was moderated by muscle dysmorphia-related attitudes, such that the association was strongest among individuals with elevated muscularity-related concerns. No comparable effects were observed for visual body size perception. These findings highlight the importance of considering embodied processes in research on body image disturbances.

Name Student: Jasha le Mair

Internship or Thesis: Internship

Title Project: Internship at Waag Futurelab Amsterdam

Abstract:

Waag Futurelab is an organization that engages in a broad range of research and development activities working towards a future of technology that is social, just and inclusive. My activities as an intern were at their research department 'CODE' and consisted of assisting in three separate projects as well as doing a literature study. Firstly, I assisted in an European initiative called 'Digit-Able' that focuses on strengthening the digital capabilities of social enterprises and their vulnerable target-groups. Secondly, I represented Waag at the Dutch Design Week for their project 'Connection Lost' about the future of social media. Lastly, I worked on a long-term research program called 'Lokale Verbindingen' about closing the digital divide in Amsterdam. Next to this I also conducted an individual literature study about an enactive cognitive psychology the attention economy and presented about this at an organization-wide meeting.

Name Company: Waag Futurelab

External Supervisor: Tessel van Leeuwen

UU Supervisor: Stella Donker

Name Student: Julie Anna Tlustá (8699402) – j.a.tlusta@students.uu.nl

Internship or Thesis: Thesis

Title project: Assessing Wayfinding Tendencies Survey in Individuals with Visual Impairment

Name company or University: Department of Experimental Psychology, Utrecht University

Assessor: PhD Candidate Dominique Blokland (MSc/MA) - d.blokland@uu.nl

Objective The thesis aims to refine the items of the Wayfinding Tendencies Survey to assess personal sensory navigation in visually impaired persons (VIP) prior to pilot testing.

Background Orientation and Mobility (O&M) training tailors instruction to individual wayfinding preferences, yet no standardised and validated instrument exists to evaluate these (Wiener et al., 2010).

Method Six participants, both VIP and O&M trainers, will complete the survey using cognitive interviews with a think-aloud protocol (Blair et al., 2010). Item-level issues will be analysed using Tourangeau's four-step cognitive response model, alongside usability testing and respondent fatigue identification (Tourangeau, 1984; Willis, 2015; Jeong et al., 2023).

Results The study identifies problematic survey items, the stages of cognitive response processing where difficulties arise, and issues related to the survey length and administration.

Conclusion The findings show how VIP experience and interpret the Wayfinding Tendencies Survey, and how these insights can guide item refinement before pilot testing.

Application By refining this survey, O&M instructors can systematically tailor training to clients' individual preferences, increase training success and thus contribute to the improvement of clients' quality of life (Blokland et al., 2026).

Grids vs. Scales: operationalizing valence and arousal.

Creating and piloting an emotion recognition task.

Anne Groot - 7079087

Internship – 12,5 ECT

Utrecht University

UU supervisor: Roy Hessels

Abstract

Aim - This internship consisted of two projects: a collaboration assignment, for which we had to provide the Dutch translation, and subsequently conduct an existing cross-cultural experiment with Dutch university students on the perception of body motions. Additionally, I replicated this experiment in Gorilla to practice with experimental building and design.

Approach - The goal of the experiment was to build a functioning online emotion recognition experiment, add two methods for measuring valence and arousal (the valence and arousal scales and the Valence-by-arousal grid) and test their validity.

Results - As expected, the results showed that both methods accurately measure valence and arousal, and that the results of the Valence-by-arousal grid are slightly more precise. Near the end of the internship, the Dutch data collection was concluded as well.

Wearable eyetracking and gazemapper manual

Wessel ten Berge

Supervisor: Roy Hessels

Wearable eye tracking is increasingly used in research, but learning how to work with it in a meaningful way remains difficult. During this internship, a manual was developed for wearable eye tracking using ArUco markers and GazeMapper. The course is structured around a single running example, a chess decision making task, and follows the full eye tracking pipeline from idea and question formulation, through experimental design and physical setup, to data processing and interpretation. Rather than treating these steps in isolation, the course emphasizes how early choices influence later ones, and how each transformation of the data introduces assumptions and limitations. By supporting readers through the complete pipeline and repeatedly linking decisions back to the original research question, the course aims to support better use of wearable eye tracking in research settings.

Name Student: Tijs van Schaik

Internship or Thesis: The World is Your Lab

Title project: Valence, Arousal, and Affiliation in Emotion Perception

(Abstract, 100-150 words)

This study aims to research the recognition of emotions in body-movements. Emotions are conceptualized within a circumplex model defined by the dimensions of valence and arousal. With valence ranging from negative to positive and arousal from low to high. The concept is based on the James–Lange theory, which emphasizes arousal as a primary determinant of emotional experience, the present study examines whether valence predicts perceived affiliation (construct based on the friendliness-hostility scale) and whether this relationship is moderated by arousal. Emotional stimuli were selected from the Asian Emotional Body Movement Database (DIEM-A), consisting of Asian performers enacting emotionally evocative situations. Participants rated each performance on perceived affiliation, as well as independently on valence and arousal. Prior to hypothesis testing, a pilot experiment using a within-subjects design to compare two answering formats for measuring valence and arousal: a slider and a circumplex grid (9×9). Results indicate that the sliders provided the most sensitive and favorable measurement of valence and arousal. Data-analyses supported the hypothesis that valence significantly predicted affiliation. However arousal is not moderating this relationship. Limitations of the circumplex approach are discussed, including potential cultural influences and the missing elements of additional emotional components such as cognitive appraisal.

Name company or University: Utrecht University

UU Supervisor: Roy Hessels

Name Student: Maria Balaska

Internship or Thesis: Internship, “The World is Your Lab”

Title project: “Got the Message?: The Role of Contrast and Brand Placement in Perceived Advertisement Effectiveness”

An advertisement must capture attention and be remembered in order to be effective. Previous research suggests that contrast levels and the placement of brand elements influence attention and memory, contributing to the perceived effectiveness of an advertisement. The aim of this study is to investigate how contrast levels (high vs. low) and logo placement (centre vs. corner) affect the perceived effectiveness of poster advertisements. To address the research objectives, 31 participants engaged into a viewing task of 12 advertisement posters, followed by an unannounced recognition test. Recognition performance was analysed using nonparametric statistical tests. The findings indicated that both contrast level and logo placement were associated with differences in recognition performance, as well as a significant interaction between the two. Brands with high contrast posters were recognised more accurately than low contrast posters. Regarding the placement of the brand, the posters with centred brand logos improved recognition primarily for low contrast posters, indicating that placement plays an important role when contrast is reduced.

Name company or University: Utrecht University

(If applicable) External supervisor: -

UU Supervisor: Roy Hessels

Name Student: Cai-E Kastelein

Internship or Thesis: Internship

Title project: "Implicit Bias of Autism in Dutch Women"

Abstract

Autism has long been framed through theories emphasising "masculine" traits, contributing to gender bias in diagnosis and the underrecognition of autistic women. Cultural expectations that women should be empathetic and socially adept often lead autistic women to mask their traits. Although autism was once thought to affect men four times more often than women, recent research suggests a ratio closer to 3:1. Prior work shows that autism is both implicitly and explicitly associated more strongly with men.

Building on their bachelor's thesis that replicated the explicit-bias component of Brickhill et al. (2023), this internship project focused on implicit bias in the perception of autism in Dutch women. Using reaction-time measures, the experiment assessed how quickly participants associated male or female names with autistic versus non-autistic traits. During the research process, features of the experimental design were identified that may influence responses, shifting the emphasis toward evaluating the method rather than testing hypotheses.

Name company or University: Utrecht University

(If applicable) External supervisor:

UU Supervisor: Roy Hessels

Bibliography

Brickhill, R., Atherton, G., Piovesan, A., & Cross, L. (2023). Autism, thy name is man:

Exploring implicit and explicit gender bias in autism perceptions. *PLoS ONE*, 18(8), e0284013. <https://doi.org/10.1371/journal.pone.0284013>

Napolitano, A., Schiavi, S., La Rosa, P., Rossi-Espagnet, M. C., Petrillo, S., Bottino, F., Tagliente, E., Longo, D., Lupi, E., Casula, L., Valeri, G., Piemonte, F., Trezza, V., & Vicari, S. (2022). Sex differences in autism spectrum Disorder: diagnostic, neurobiological, and behavioral features. *Frontiers in Psychiatry*, 13. <https://doi.org/10.3389/fpsy.2022.889636>

The Adaptation of Digital Learning Platforms: Brightspace Pulse at Utrecht University

Abstract

This study focuses on the factors that contribute to the adaptation of the digital learning platforms of Utrecht University, especially (Brightspace) Pulse, and how to improve its usage and awareness. The Pulse app seems to lack in those factors as a mobile app, despite it having the same functionality and course related support as the desktop version. For this study, a quantitative research method was conducted based on an online questionnaire (N=39) with adaptation of the System Usability Scale (SUS). To clarify this data, a qualitative interview method was used (N=5). Results indicate that the low awareness of Pulse was caused by low prominence, indistinct communication about its existence, and uncertain benefits to the student's study routine. Furthermore, the results demonstrate that usage rates could be enhanced by improved communication and engagement strategies. Considering these findings, the study recommends improved integration and clearer communication by Utrecht University about Pulse.

Key words: Brightspace Pulse, Utrecht University, Digital Learning Platform, System Usability Scale, MyUU

Name Student: Evita van Klaren (9397795)

Internship or Thesis: Internship

Title project: Cognitive Load and Workflow Analysis of Participant Scheduling Practices in UX Research at Capgemini

This exploratory study investigates how UX researchers at Capgemini organize and execute participant scheduling for qualitative research sessions and examines how workflow inefficiencies affect cognitive load and perceived efficiency. Although participant scheduling is a core component of UX research operations, it remains largely underexplored despite its potential impact on research quality and efficiency. A mixed qualitative approach was used, combining a short survey (N = 8) with semi-structured interviews (N = 5). The findings indicate that participant scheduling is managed through fragmented, manual workflows across multiple tools, leading to repetitive coordination, loss of overview, frequent task interruptions, and increased cognitive load. These inefficiencies were reported to disrupt other research activities and reduce researchers' perceived productivity. The study concludes that participant scheduling constitutes a structural workflow challenge and identifies opportunities for integrated and automated support to reduce cognitive demands and improve UX research operations.

Name company or University: Capgemini

External supervisor: Christina Veelenturf (christien.veelenturf@capgemini.com)

UU Supervisor: Marnix Naber

Name student: Marin van Meezen

Internship or Thesis: Thesis

Title project: Locating Numerosity Adaptation in the Visual System: An Interocular Transfer Study.

Perceiving numerosity, the representation of non-symbolic quantities, is fundamental for cognitive functions. Previous research into numerosity show contradicting objectives on the neural locus of this process. It is unclear whether numerosity processing emerges as a subcortical or cortical process. The current study aims to provide insight into this debate by examining numerosity using adaptation in an interocular transfer (IOT) task. Adaptation to visual stimuli cause aftereffects, which in turn provides a window into visual encoding of perceptual abilities such as numerosity. The dichoptic viewing experimental set-up ensures manipulation of stimulus, allowing both monocular and binocular stimulus presentation. Interocular Transfer (IOT) occurs when adaptation-effects of stimuli presented to one eye are still present for stimuli presented to the other. If numerosity adaptation primarily reflects subcortical mechanisms, no interocular transfer should occur. In contrast, partial or full interocular transfer suggests numerosity as a higher-order, cortical representation.

Name company or University: University of Utrecht

UU Supervisor: Prof. Dr. Maarten van der Smagt

Estelle Coelingh
- Internship

Title:

The Distracted Brain: Tracking Thoughts to Words with Working Memory

Abstract:

Everyday communication requires speakers to maintain their own intended utterances while continuously updating information from others. Despite this complexity, most experimental research investigates language comprehension or production in isolation. This study examines how working memory updating and maintenance interact during speech production under distraction. Participants perform a dual-task language paradigm in which they name pictures while processing sentence contexts that vary in sentence-target congruency and length. Congruency manipulates working memory updating demands, while sentence length manipulates maintenance demands. Behavioral responses and electroencephalography (EEG) are recorded to assess naming performance and neural oscillatory activity. Preliminary behavioral results show differences in naming reaction times across conditions, while EEG analyses are ongoing. This research aims to provide insight into the neural mechanisms that support fluent speech production under cognitively demanding conditions.

Leiden University

External supervisor: Dr. Xiaochen Zheng

Internal supervisor: Dr. Leon Kenemans

Name Student: Mathieux Panacho

Internship or Thesis: Internship

Title project: Improving viewing behaviours of new racing drivers through viewing instructions – a first insight created with eye tracking

(Abstract, 100-150 words)

This paper examines how viewing instructions influence professional racing drivers' visual behaviour and on-track performance. Drivers typically learn to look further ahead over time, but acquiring this skill earlier may enhance both safety and performance. The study aims to provide initial insight into whether targeted viewing instructions can accelerate improvements in gaze behaviour and performance metrics. Using a racing simulator, participants' eye movements were recorded with a wearable eye tracker and analysed via Pupil Player and manual annotations, alongside simulator telemetry data. Post-session verbal questions were used to capture drivers' subjective experiences. Results indicate that viewing instructions led drivers to look further ahead on the track, which was associated with improved performance measures. The findings suggest that structured visual guidance can positively affect both visual strategies and driving outcomes. The study concludes that further research in real-world racing environments is needed, as these insights could eventually be applied in professional racing schools.

Name company or University: Utrecht University – The World is Your lab

(If applicable) External supervisor: -

UU Supervisor: Roy Hessels

Name Student: Isabela Fernanda Vilanova Sanchez

Internship or Thesis: Thesis

Title project: Error Monitoring, Cognitive Control, and Irrelevant Advice in Human–AI Decision Making

Abstract

Despite the increasing accuracy of artificial intelligence (AI) systems, people often show inconsistent reliance on AI advice, including over-reliance or algorithm aversion following errors. Most prior research explains these behaviors in terms of trust or attitudes. The present study proposes a complementary cognitive account, examining advice-taking through the lens of error monitoring and cognitive control. Approximately 80 adult participants will complete a perceptual decision-making task involving ambiguous face–car images while receiving advice from either an AI system or a human advisor. Advice accuracy is experimentally manipulated, and participants’ choices, reaction times, confidence, and advice adherence are measured across repeated trials. Using linear mixed-effects models, the study investigates whether advisor type modulates behavioral indices of conflict and adjustment following errors. By reframing algorithm aversion as a performance-monitoring phenomenon, this research aims to provide mechanistic insight into how humans integrate or resist AI input and to inform the design of AI systems better aligned with human cognitive control processes.

Name company or University: Utrecht University

UU Supervisor: Baptist Liefoghe and Leendert van Maanen

Name Student: Duncan Uittenbogaard

Internship or Thesis: Internship

Title project: Kickstart your wearable eye-tracking project (KET)

Wearable eye-trackers can look like the ultimate solution for your project. However, using wearable eye-trackers is more complicated than one may think, which is where this internship project comes in. Together with Wessel ten Berge, I made a small course on how to use a wearable eye-tracker and a small chess experiment to function as an example through the course. The course consists of four parts and each part highlights different challenges of eye tracking. The first part will focus on whether a wearable eye-tracker fits your experiment or project at all. The second part will focus on building an actual experiment and what things you will have to consider. The third part will be dedicated to the basic functionality of gazeMapper, which is an incredibly useful tool for transforming eye-tracking data. The fourth part will be about data-analysis and being able to draw conclusions about eye-tracking data.

Name company or University: Universiteit Utrecht, "The world is your lab"

UU Supervisor: Roy Hessels

Name Student: Kamiah Metz

Internship or Thesis: Internship

Title project: Aligning with Practice: Optimizing the Consult Preparatory Questionnaire for Primary Mental Health Professionals

Abstract

The POH-GGZ works in an increasingly busy and complex primary care setting, leading to a growing need for smart support. DigiDok offers a consult-preparatory questionnaire to support the first consultation; however, it is unclear whether this questionnaire aligns with daily practice and whether additional digital tools are needed. To investigate this, the perspectives of POH-GGZ professionals were collected through semi-structured interviews. This explored opportunities to optimize the current questionnaire and whether digital support could address existing challenges.

The results show that the questionnaire is mainly used as a conversational aid rather than as true preparation for the consultation. In addition, several content-related and practical points for improvement were identified to better align with clinical practice. No clear need for new digital tools emerged. The findings suggest that strengthening the existing questionnaire is a more appropriate and feasible approach. The recommendation focuses on repositioning the questionnaire, simplifying its content, and optimizing preconditions to improve its fit with POH-GGZ practice.

Name company: DigiDok

External supervisor: Emmely Pennings

UU Supervisor: Larissa Meijer

Name: Kateřina Kozlová

Internship or Thesis: Internship

Title project: Gender Differences in Perception of Friendly/Hostile Movements

Name company or University: Utrecht University

UU Supervisor: Roy Hessels

Body movement databases and studies of emotion perception often overlook cultural and gender context. This study presents a replication based on the DIEM-A movement database, introducing a European perspective through a Dutch participant sample. The original DIEM-A questionnaire was translated into Dutch, and participants completed an online task in which they evaluated point-light display animations depicting body movements performed by Japanese and Taiwanese actors. Movements were rated on a continuous scale ranging from 0 (friendly) to 100 (hostile). Data analyses explored potential gender differences in perceived friendliness and hostility. Due to the small and gender-imbalanced sample, results are reported without drawing conclusion; however, observed patterns suggest subtle gender-related differences consistent with previous emotion perception research. The findings highlight the importance of culturally diverse samples when examining emotion perception from body movement.

Iason Balomas

Internship: The World Is Your Lab (12,5 ECTS)

Title Project: The Influence of Background Music In Online Shopping Decisions

Abstract

This study examined whether background music tempo and familiarity influence consumer behavior in an online clothing shopping context. 35 participants browsed an online store under four music conditions: familiar, unfamiliar, low-tempo, and high-tempo music. Emotional, cognitive, and self-reported behavioral measures were collected. Results showed that familiar music was associated with more positive mood, while low-tempo music increased emotional energy, supporting findings from previous retail research. However, effects on actual purchase behavior were inconclusive due to methodological limitations. Behavioral outcomes relied on retrospective self-reports rather than real-time, condition-specific data, limiting the ability to link effects to individual music conditions. This study demonstrated that empirical research requires precise alignment between theory, measurement, and analysis. It highlighted the importance of collecting real-time behavioral data, creating conditions clearly, and selecting appropriate statistical methods to ensure that research questions are accurate.

Utrecht University

Supervisor UU : Roy Hessels

Name Student: Matilda Kühme

Internship

Utrecht University: The World is Your Lab

UU Supervisor: Roy Hessels

Seat-Finding in the Utrecht University Library: Evaluating Wayfinding Effectiveness and Efficiency for First-Time Users

For first-time users, finding a booked seat in a library can be cognitively demanding, especially when signage and spatial cues are ambiguous. This study evaluates the effectiveness and efficiency of the seat-finding process at the Utrecht University Library City Centre for first-time users. Seven participants with no prior experience with the booking system were asked to find four pre-selected seats while being observed. Wayfinding performance was assessed by calculating the success rate and relative navigation inefficiency. The results revealed that, in 70% of cases, participants successfully found their seat, and when they did, it took them 2.14 times longer to locate their seat than the optimal time, which varied depending on the difficulty of the seat. Misleading and insufficient signage was identified as the main issue. Therefore, improving the signage, especially at important decision points, is recommended.

Name Student: Evita van Klaren (9397795)

Internship or Thesis: Internship

Title project: Cognitive Load and Workflow Analysis of Participant Scheduling Practices in UX Research at Capgemini

At Capgemini, UX researchers currently coordinate participants for live research sessions using a fragmented set of tools within the Microsoft 365 environment, which may introduce inefficiencies and increased cognitive load. This exploratory study examines how UX researchers organize and execute participant scheduling, identifies key workflow bottlenecks, and explores the cognitive demands associated with these activities. A mixed qualitative approach was employed, combining an exploratory survey with semi-structured interviews. Descriptive survey results and thematic interview analysis revealed that scheduling is time-consuming, highly repetitive, and mentally demanding due to manual coordination, tool fragmentation, and uncertainty in participant availability. Researchers reported frequent interruptions and reduced focus on core research tasks. The findings highlight clear opportunities for workflow optimization and automation to reduce cognitive load and improve research efficiency.

Name company or University: Capgemini

(If applicable) External supervisor: Christina Veelenturf

UU Supervisor: Marnix Naber

Name Student: Daidipya Mane

Internship or Thesis: Internship

Title project: The Attention Blueprint: What Makes Creative Assets Actually Work

This report enhances Brandpulse Analytics' creative asset pre-testing service by enabling the company to explain *why* client assets capture attention, not just *which* ones perform best. Previously, clients received test results without actionable insights for future creative development. So, through a systematic review of neuroscientific and psychological literature on bottom-up attention, 17 key visual features were identified as robust attention-drivers. Therefore, this report is a structured framework containing: (1) the attention-driving features, (2) evidence-based explanations of how each influences attention, (3) visual examples of each, and (4) supporting literature references. Recommendations for improving pre-testing methodology and statistical analysis were also provided. This framework allows Brandpulse to transform client reports from simple performance rankings into science-backed recommendations, extending value beyond individual projects by enabling clients to optimise future creative assets independently.

Name company or University: Brandpulse Analytics B.V.

(If applicable) External supervisor: Arno Kooijman (arno@brandpulse-analytics.com)

UU Supervisor: Dr. Daan van Rooij

Name Student: Evy Blok

Internship or Thesis: Internship

Title project: Designing digital game interventions to reduce intrusive memories after a traumatic experience

Intrusive memories are a core symptom of acute stress disorder and post-traumatic stress disorder, yet early, scalable interventions remain limited. Visuospatial digital game interventions, such as Tetris, have shown promise in reducing intrusive memories by interfering with memory consolidation and reconsolidation processes. This systematic review examines which design features of visuospatial digital games contribute most effectively to reducing intrusive memories following traumatic experiences. A structured literature search identified 12 empirical studies investigating game-based visuospatial interventions after analogue or real-life trauma. Studies were synthesised based on five key design features: intervention timing, gameplay duration, visuospatial load, interaction type, and delivery format. Findings suggest that early intervention, sufficient gameplay duration, high visuospatial working memory engagement, active interaction, and accessible delivery formats are essential for effectiveness. The review concludes with practical design recommendations and directions for future research to support the development of scalable, user-centred digital interventions for trauma-related intrusive memories.

Name company or University: Utrecht University

(If applicable) External supervisor: -

UU Supervisor: Roy Hessels

Name Student: Jannis Klaus Hollstein

Internship or Thesis: Internship

Title project: Designing Conversational Paradigms to Study Deception and Attention in Triadic Interaction

Face-to-face conversation relies heavily on nonverbal cues such as gaze direction and head orientation to regulate attention and social interaction. Studying these processes becomes particularly challenging in contexts involving deception, where interaction should remain natural while still allowing systematic analysis. This internship focused on the design and evaluation of a conversational experimental setup suitable for studying deception and attention in triadic interaction under practical laboratory constraints. Two candidate paradigms were developed and piloted: *Two Truths and One Lie* and the *Imposter Game*. These paradigms were evaluated with respect to conversational flow, clarity of deceptive events, participant engagement, and feasibility when using only two eye trackers. Pilot observations indicated that structured tasks with explicit decision moments supported fluent interaction while enabling interpretable analysis. Based on these findings, *Two Truths and One Lie* is recommended as a particularly suitable paradigm for future multimodal studies of deception in triadic conversations.

Name company or University: Utrecht University

UU Supervisor: Roy Hessels

Name Student: Hidde Tjepkema

Internship or Thesis: Internship

Title project: Air Traffic Control Selection and the Reliability of the Visuo-spatial test (Mouratille et al., 2026) measuring Trait and State

Air traffic control is a highly complex, demanding, and responsible profession. The costs of training future air traffic controllers are very high, and the base rate of suitable candidates is low. Air traffic controllers have great responsibilities because every choice involves lives. Therefore, it is important to select the most suitable candidates. Cognitive ability tests are a standard part of the selection process for future air traffic controllers. Although the cognitive ability tests are taken on a single day, they must be generalizable and not dependent on the individual's state during test administration. This report presents insight into the selection process and predictors for future air traffic controllers, and a replication of the visuo-spatial test from Mouratille et al. (2026). The complete replicated experiment is described, including considerations because of the incomplete description of methods for the original experiment.

Name company or University: Utrecht University

UU Supervisor: Roy Hessels

Name: Tristan Zieltjens

Internship

Title project: The Psychology Behind Data Visualization at CGI

Within the CGI iAMLAB team, a recurring problem is that data visualizations, such as dashboards and presentations, are often created without a scientific foundation. This can lead to misinterpretation of data and reduced client trust. This internship aimed to address that gap by developing a manual based on psychological and cognitive science, providing evidence-based guidelines for more effective data visualization. The scope of this manual includes principles related to attention, memory, client trust, color psychology, cultural differences, and accessibility, supported by visual examples from CGI dashboards and presentations. To research this, a literature review was conducted, together with interviews with iAMLAB developers and clients. By applying these scientifically proven techniques, the manual serves as a practical resource for CGI iAMLAB developers to improve clarity, engagement, and trust in data-driven communication.

Name company: CGI Nederland

Name external supervisor: Susanne Pechtold (susanne.pechtold@cgi.com)

UU supervisor: Sjoerd Stuit

Name: Hedi Blom

Internship or Thesis: Internship

**Title: Differentiation Strategy for User Sense in the European UX Research Market:
Understanding Client Needs and Competitive Positioning**

This qualitative study examines how professionals in the European UX research market experience and make sense of digital user research platforms, with the aim of informing differentiation strategies for User Sense. Semi-structured interviews were conducted with four participants from commercial, product, and academic contexts who had hands-on experience with user research platforms. Data were analysed using reflexive thematic analysis following Braun and Clarke's approach. Six themes were identified, including research as a continuous practice, the cognitive and emotional load of analysis, trust and confidence in insights, collaboration and translation, cost awareness, and ambivalence toward AI-supported features. The findings show that research platforms are not experienced only as technical tools, but as aids that help users make sense of research under time pressure and organisational constraints. The study suggests that competitive advantage may come from helping users feel confident in their insights, understand how results are produced, and interpret findings together, alongside efficient functionality.

Name company: User Sense

(If applicable) External supervisor: Tim van der Gouwe

UU Supervisor: Daan van Rooij

Art Perception in Lindau Crucifixion Abstract

This study investigates perceptual differences between two versions of the Lindau Crucifixion painting, painted by Master Lindau in the fifteenth century. Originally, the painting had a gilded background color; the background was later painted with a dark blue color. Raising questions whether the blue color should be removed or not, these questions and discussions led to investigate the perceptual differences of the versions. Previously, a computer based study was conducted with two versions of the paintings on visual perception with eye tracking and gist perception experiments. For the current study, we have conducted the same tasks with physical facsimiles in a museum like environment. During the eye tracking task, faces on the painting were consistently gazed at, while angels were gazed at less. However, in the blue background version, it is found that participants gazed at angels much less than in the gilded version. In the gist perception task, participants described central figures of the painting regarding the background color; however, a slight difference on the gist perception of the angels were found. These findings suggest perceptual differences between two versions of the same paintings and highlight the potential of perceptual experiments to inform art conservation decisions.

Keywords: art perception, eye-tracking

Name: Jeanne Gaessler

Internship or Thesis: Internship

Title project: Creating an accessible website that critically compiles research on dyslexia and dyslexia-friendly fonts.

Abstract: Typotheque is a typography foundry based in The Hague, and is often inquired by prospective clients about dyslexia and its relation to type design. Currently, there is a lack of online resources that compile research and other relevant information on what dyslexia is, its impact in different languages, and (research on) dyslexia-friendly fonts. Thus, the aim of this internship was to create an accessible and organised digital resource that critically compiles the latest research and information on dyslexia and dyslexia friendly fonts. A specific sub-goal was to write it so that it could be understandable for readers who do not come from an academic/psychological background. To achieve this, I created mock-ups for the website's structure and wrote up the text in markdown files. The final product will be a website (accessibility.tools) that can be shared with Typotheque's prospective clients and any other interested readers.

Name of company: Typotheque

(If applicable) External supervisor: Héctor Mangas

UU Supervisor: Anouk Keizer

Name Student: Alva Sihvo

Internship or Thesis: Internship 12.5 ECTS

Title project: Can We Use Pupillometry in Infants? Assessing Data Usability and Loss in an Infant Language Learning Paradigm

Pupillometry is increasingly used to study cognitive and attentional processes, yet its feasibility and data quality in infant populations remain insufficiently documented. This study examines whether pupillometry can be reliably used with 18-month-old infants by evaluating the amount of usable pupil data obtained per infant and the proportion of data loss during an experimental language learning task. By combining pupillometry with the Head-turn Preference Paradigm (HPP), this research provides methodological insights into the practical constraints and reliability of pupillometric measures in infant research settings. An experimental design was employed in a Babylab laboratory with 18-month-old Dutch infants ($N = 34$). Pupillometry data were collected alongside HPP measures and analyzed using Python DataMatrix to quantify usable pupil samples and sources of data loss (e.g., movement artifacts and calibration failures). Results showed variability in usable data across infants, with significant data loss primarily due to movement. Overall, pupillometry proved feasible but subject to practical constraints.

Name company or University: Babylab for Language Research at Utrecht University

(If applicable) External supervisor: Areti Kotsolakou

UU Supervisor: Stella Donker

Name Student: Keying Chu (7716516)

Internship or Thesis: Internship

Title project: Investigating Art Perception Through Physical Facsimiles: Lindau Crucifixion

Abstract: This study aims to investigate how background color influences viewers' general perception of the Lindau Crucifixion, an early fifteenth-century panel painting attributed to the Master of the Lamentation of Christ in Lindau. It was discovered that the painting's current dark-blue background is a later intervention, whereas the original background was gilded. This discovery raises a dilemma concerning whether to preserve the later overpaint or to restore the original gold background. To explore the potential perceptual consequences of this restoration decision, the researcher uses high-quality physical facsimiles of the painting in perception experiments. By integrating methods from cognitive psychology with questions from art conservation, this study seeks to provide empirical evidence that may inform restoration decision-making and contribute to interdisciplinary dialogue between perceptual science and heritage conservation.

Name company or University: Utrecht University

UU Supervisor: Roy Hessels

Karoline Ruppert
Internship project „The world is your lab“

Studying deception, attention, and eye-head coordination in triadic conversations

Abstract

Understanding gaze behavior in social interaction requires reliable measurement of head orientation in addition to eye-tracking, particularly in multi-person, face-to-face conversations. This internship project focused on the development, piloting, and technical refinement of a video-based head-tracking setup designed to capture head orientation during conversational interaction. Head movements were recorded using a ceiling-mounted camera in combination with ArUco fiducial markers attached to participants' helmets in a triadic interaction setting. The primary contribution of this report lies in the systematic identification of technical challenges encountered during piloting, including camera configuration, marker placement and visibility, body movement artifacts, and issues in data preparation and processing. For each challenge, practical solutions are described and illustrated, along with validation procedures confirming reliable capture of horizontal head orientation. The report is intended as a practical reference for future users, outlining common problems and effective solutions for implementing robust head-tracking setups in naturalistic social interaction research.

Utrecht University
UU Supervisor: Roy Hessels